

## Feature 5 (Quiz options):

**Problem:** The quiz currently generates a fixed 5 questions regardless of how much material has been uploaded. For students who have indexed large amounts of content, multiple PDFs, full lecture slide decks, a 5-question quiz provides superficial coverage and cannot meaningfully assess understanding across the full material. Conversely, students who do a quick review before a short-topic quiz may find even 5 questions excessive. The fixed count serves neither use case well.

**Target students:** Any student who has indexed material and wants to take a quiz, but specifically students with large indexed workspaces where 5 questions leave most topics untested.

**Proposed solution:** Let students select the number of questions before generating the quiz. Provide preset options (5, 10, 15, 20) with a recommended default based on the amount of indexed material. The recommended default is calculated based on the number of sections in the generated study guide, with approximately 2 questions per section, capped at 20.

### Goals:

- **Primary metric:**
  - The average number of questions selected per quiz session exceeds 5, indicating that students want more coverage than the current default provides.
- **Secondary metric:**
  - Quiz completion rate does not decrease as question count increases. If students are abandoning longer quizzes at a high rate, the upper limit needs to be adjusted.
- **Guardrails:**
  - Quiz generation time does not increase beyond user tolerance. More questions mean a larger prompt. Monitor generation time by question count and add a warning if generation exceeds 30 seconds for high counts.

## MVP / Functional Requirements

### Core flow:

- [P0] Count respected in generation: The quiz generation prompt explicitly requests the selected number of questions and validates that the returned quiz JSON contains the correct count before rendering.

[P1] Study mode integration: Cram Mode defaults to 5 questions. Deep Dive mode defaults to the material-based recommendation. The student can override either default manually.

- [P1] Content volume floor: If the indexed material has fewer than 3 sections, the maximum selectable count is 10. Generating 20 questions from a single-page upload produces repetitive or low-quality questions. Show a tooltip explaining why higher counts are unavailable.
- [P2] Count shown on results screen: After completing the quiz, the results screen shows how many questions were answered out of how many were generated, and what percentage of study guide sections were covered. This gives students a sense of how comprehensive their review was.

## Edge Cases

- The student selects 20 questions, but has only one source file with 2 sections, handled by the content volume floor in P1. The maximum available count is reduced, and the student is informed of the reason.
- If the student selects 20 questions, and the AI returns 18 due to prompt limitations, the app should not crash or show a broken results screen. Accept what is returned and show "18 of 20 questions generated" on the results screen.
- Gemini hits a 503 during a 20-question generation; this is more likely with larger prompts. The existing generation\_failed event should fire, and the user should see a specific message rather than a generic error.
- The student is in Cram Mode but manually selects 20 questions, allowing the student to do so. The mode default is a suggestion, not a restriction.
- If weak topics exist, a "Generate study guide from weak areas" option appears below the quiz results.
- [P0] On click, the system retrieves indexed material chunks limited to the tagged weak topics.
- [P0] The system generates and displays a study guide scoped to those weak topics only.

## Out of Scope:

Unlimited custom input (student types any number). Per-topic question allocation (3 questions on topic A, 2 on topic B). Adaptive question count that changes based on past performance (future consideration for spaced repetition system).